

Symbiosis Institute of Technology, Pune

DCDS Lab ESE Rubrics

Department: Artificial Intelligence and Machine Learning

Batch: 2025-29

Course name: Database Concepts for Data Science Lab

Credit: 1

Year / Sem: SY /Sem 3

Name of the faculty member: Dr. Anjali Dalvi, Dr. Sumanto Dutta,
Dr. Wasim Khan

Sr. No.	Component	CO	Max marks	Weight	Tentative date
1	ESE - Miniproject Python-Based Data-Driven Application Development with Advanced with SQL and NoSQL Integration	CO1, CO2, CO3, CO4, CO5	10	40 %	26 to 30th October 2026

ESE Rubrics:

Python-Based Data-Driven Application Development with Advanced with SQL and NoSQL Integration – 10 marks

Evaluation Criteria	Excellent	Good	Satisfactory	Needs Improvement	Max Marks
Application Design & Functionality	2.0	1.5	1.0	0.5	2
Python Programming Implementation	2.0	1.5	1.0	0.5	2
Database Integration (SQL & NoSQL)	2.0	1.5	1.0	0.5	2
Advanced Database Querying & Data Retrieval	2.0	1.5	1.0	0.5	2
Code Quality, Documentation	1.0	0.75	0.50	0.25	1

& User Interface					
Testing, Validation, Security & Error Handling	1.0	0.75	0.50	0.25	1
Total	10	7.5	5.0	2.5	10

ESE: Expected functionality under each evaluation parameter

Evaluation Criteria	Excellent	Good	Satisfactory	Needs Improvement	Marks
Application Design & Functionality	Fully functional application meeting all requirements with intuitive workflow and proper modular design.	Functional application with minor issues.	Basic functionality implemented with limited features.	Incomplete or non-functional application.	2 Marks
Python Programming Implementation	Efficient use of Python concepts such as functions, classes, exception handling, file handling, and modular coding.	Good use of Python with minor coding issues.	Basic Python implementation with limited best practices.	Poor coding practices and significant errors.	2 Marks
Database Integration	Seamless integration with SQL/NoSQL database, proper connectivity, and data manipulation.	Database integration works with minor issues.	Basic connectivity established but limited operations.	Poor or incorrect database integration.	2 Marks
Advanced Database Querying	Effective use of advanced queries (Joins, Nested Queries, Aggregations, Grouping, Stored Procedures, MongoDB Aggregation	Good use of advanced queries with minor errors.	Basic CRUD operations with limited advanced querying.	Queries are incorrect or lack complexity.	2 Marks

	Pipeline, etc.).				
Code Quality, Documentation & Output Presentation	Well-structured code, proper comments, meaningful variable names, clear output/interface, and complete documentation.	Good documentation and presentation with minor gaps.	Limited documentation and basic presentation.	Poor documentation and unclear output.	1 Mark
Testing, Validation & Error Handling	Comprehensive testing, input validation, and robust error handling implemented.	Adequate testing and validation.	Basic testing with limited validation.	Minimal or no testing/error handling.	1 Mark

